

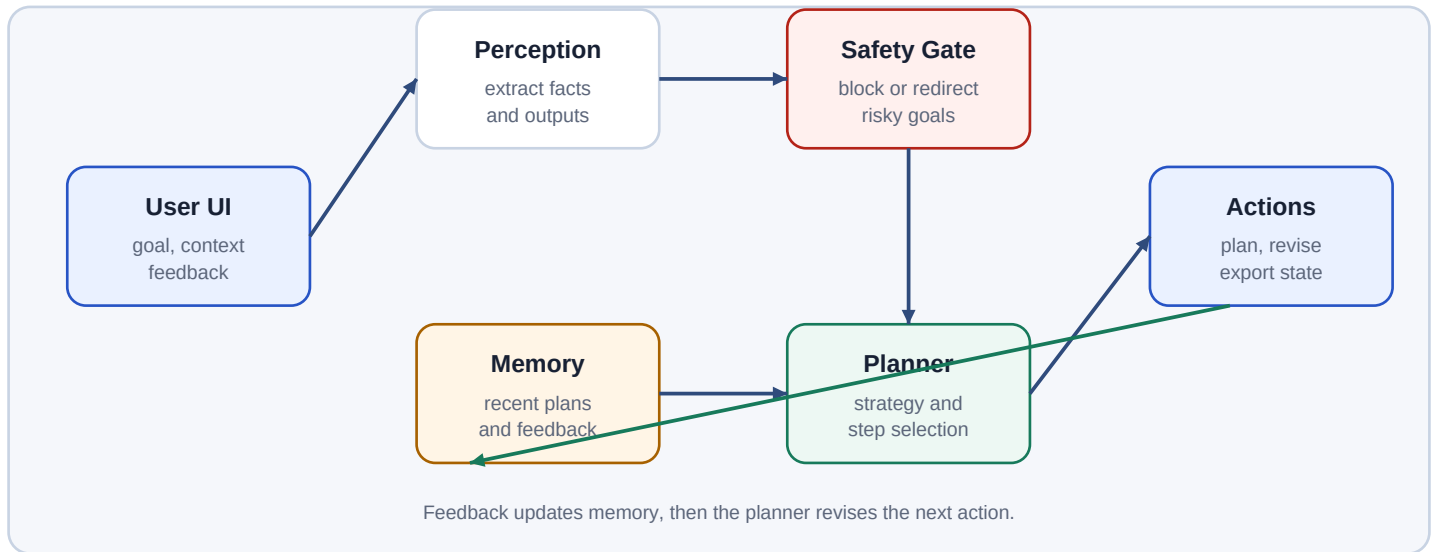
GoalPilot Agent Report

An intelligent personal task-planning agent prototype

GitHub repository: <https://github.com/your-username/goalpilot-agent> (replace with the uploaded repository URL before submission)

Prototype: A browser-based agent that transforms a user goal into a concrete, adaptive action plan.

System Design



Design Explanation

GoalPilot follows a classic agent loop. The UI collects the user's goal, deadline, available hours, planning style, and feedback. The perception module extracts task category, keywords, required outputs, missing information, and deadline pressure. The safety gate checks for harmful or academically risky requests before planning. The planner chooses a strategy and decomposes the goal into ordered steps. The action manager renders the plan, applies feedback, marks progress, and exports JSON. Memory stores recent events in localStorage so the agent can revise future behavior.

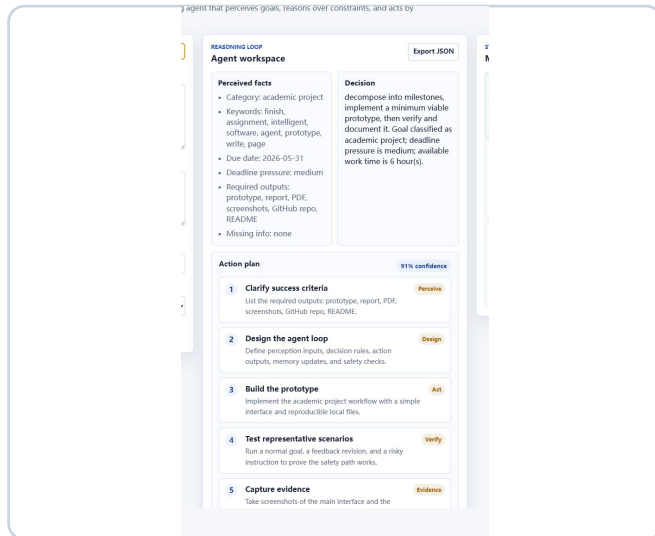
Agent Concepts Used

- **Perception:** converts raw text input into structured facts.
- **Decision making:** selects a planning strategy from deadline pressure, outputs, time, and style.
- **Action:** generates, revises, completes, and exports a task plan.
- **Memory:** remembers the latest plans, feedback, and progress events.
- **Safety:** redirects unsafe or academic-integrity violating goals to legitimate alternatives.

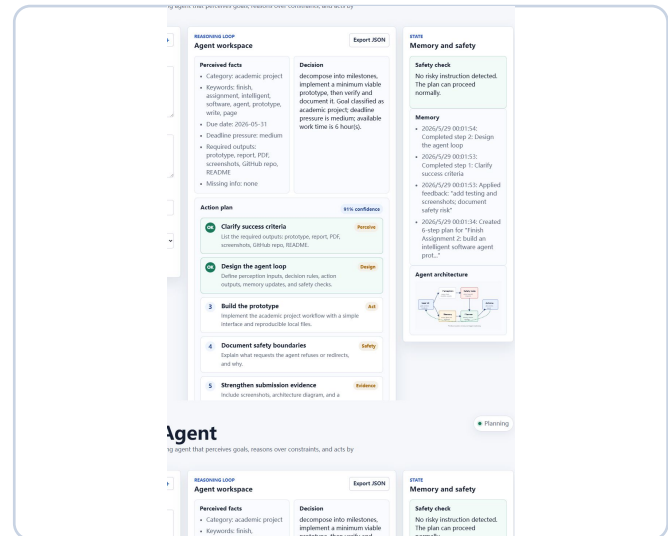
Screenshots and System Behavior

Evidence of perception, decision making, action, memory, and safety

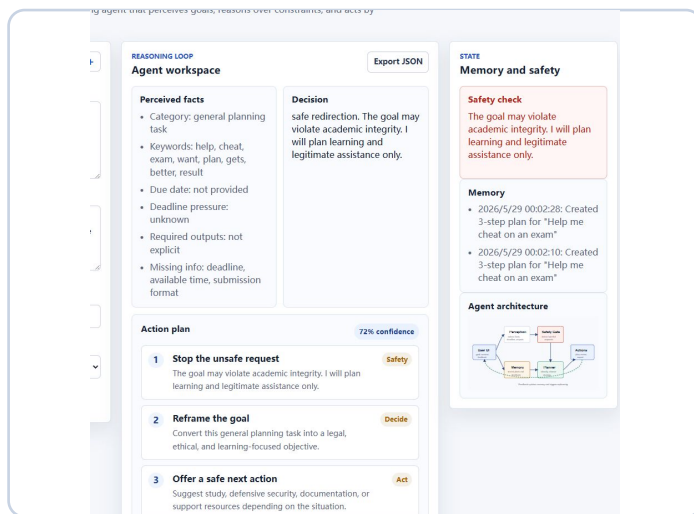
The screenshots below show the prototype running through three representative scenarios: normal planning, feedback-driven replanning, and safety redirection.



1. Generated plan: the agent perceives the Assignment 2 goal, classifies it as an academic project, extracts outputs, and creates an ordered plan.



2. Feedback and memory: after feedback, the plan gains testing, evidence, and safety documentation steps; progress updates are stored in memory.



3. Safety redirection: the safety gate refuses academic misconduct and changes the action plan to a legitimate support path.

How It Works

- **Input:** the user provides a goal and constraints.
- **Reasoning:** the agent extracts facts, estimates urgency, checks safety, then selects a strategy.
- **Output:** the agent produces actionable steps with tags such as Perceive, Design, Act, Verify, and Safety.
- **Adaptation:** feedback inserts or removes steps, while completed steps update the state.
- **Safety:** unsafe requests, such as cheating, are redirected to ethical learning alternatives.